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Design optimization of steel–concrete hybrid wind turbine tower based on improved genetic algorithm

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Summary

The steel–concrete hybrid wind turbine tower is characterized by the lower part of the traditional steel tubular tower replaced with the concrete segment. The lateral stiffness will be improved obviously, and then, the excessive vibration of the steel tower can be solved effectively. Based on the improved genetic algorithm, an optimization program is built to consider the influence of materials, labor, machinery, and transportation on the construction cost of a steel–concrete hybrid tower for a 2.0-MW wind turbine with a hub height of 120 m, in which the initial height of the concrete segment is 32 m. During the optimization process, design requirements of relevant specifications and industry standards are used as the constraints. The optimization variables include the bottom and top diameters of the tower, the wall thickness of each segment, the height of the concrete segment, and the area of the prestressed steel strand. By comparing the results of construction cost and structural capacity before and after optimization, it can be found that the steel–concrete hybrid wind turbine tower after optimization has the better structural stiffness and lower construction cost. The proposed optimization program can meet the design requirements and significantly improve the economic performance of the tower.

KEYWORDS

construction cost optimization, genetic algorithms, steel–concrete hybrid tower, structural analysis, structural optimization, wind turbines

1 | INTRODUCTION

Wind energy, which can be obtained almost anywhere, is a clean, economical, and renewable energy. Horizontal-axis wind turbine rotor mounted on the top of a tower was mostly used to generate electricity by harvesting wind energy. The common structural system of wind turbines is a tubular steel tower formed by rolling steel plates in factory and connected by flanges in situ. Generally, the cost of the tower would amount to approximately 15%–20% of the total cost for onshore multimegawatt wind turbine system. Therefore, there is a high potential to reduce the cost of the tower by structural optimization. In recent two decades, some researchers have focused on the design optimization of wind turbine towers. Takewaki^[1] developed a numerical method via an approximation concept for finding the optimal bending stiffness distribution of a tower structure with tube-type cross section under a fundamental natural frequency constraint. Negm and Maalawi^[2] presented several optimization models for the design of a typical wind turbine tower structure in which the cross-sectional area, radius of gyration, and height of each segment were chosen as the effective design variables. Significant works have been performed to focus on the design optimization of wind turbine towers. Uys et al.^[3] presented the cost optimization of a slightly conical ring-stiffened welded steel shell, in which the optimum shell thickness, number of stiffeners, and dimensions of the stiffeners were calculated using Rosenbrock's search algorithm. Based on the automatic generation of tower finite element models characterized by a few parameters, Zwick et al.^[4] presented basic considerations for an iterative optimization approach and

identified sensitivities for the optimization process of a full-height lattice tower. Quilligan et al.^[5] investigated the relative performance of steel and concrete tower solutions for a multimewatt turbine at hub heights in the range of 80–120 m. Long et al.^[6] proposed an optimal design methodology of lattice towers for offshore wind turbines in the ultimate limit state considering that tower mass and fabrication complexity. Hu et al.^[7] analyzed the effect of internal stiffening rings and wall thickness on the structural response of steel tubular wind turbine towers by FEM modeling. Stratton et al.^[8] developed a design methodology considering the relationship among the mechanical performance, cost, and environmental impact and determined the optimal tower design using a multilevel decision-making procedure. Based on an analytical gradient-based method, Chew et al.^[9] proposed an integrated optimization framework to optimize offshore wind turbine support structures in an effective and efficient way. Based on life-cycle cost measurements, Lagaros and Karlaftis^[10] proposed a design optimization procedure of steel wind towers using the differential evolution algorithm. Based on a coupled parametric finite element analysis and genetic algorithm, Gentils et al.^[11] developed a structural optimization model for offshore wind turbine support structures to minimize the mass of the support structure under multicriteria constraints. Young et al.^[12] presented a methodology for the design and optimization of a composite wind turbine tower for use on floating offshore turbines using the genetic algorithm. Oest et al.^[13] proposed three gradient-based optimization methods of jacket structures for offshore wind turbines considering both mass and fatigue constraints. Based on single-objective evolutionary algorithm-driven optimization, Bukala et al.^[14] presented a computing methodology of a guy-wired mast structure for small-scale wind turbines.

In the global wind energy market, wind farms with high-quality wind resources have been exploited progressively in recent 10 years, and more wind turbines will have to be installed in low wind speed areas. In order to obtain greater and more stable wind speeds, wind turbine towers are becoming taller and taller for supporting the larger rotors. Many wind turbine industries have tried their best to develop the taller wind turbine with longer blades and more powerful generator for producing more electricity. As early as 2017, the steel tower height of one wind turbine produced by Vestas and installed in Germany has broken through 160 m. As turbine sizes increase, steel towers are required to have an increase of thickness and diameter for ensuring sufficient strength and bending stiffness. However, the flexural stiffness of the steel tower does not increase proportionally due to the limits in manufacturing and transportation and then leads to a decrease in natural frequency.^[15] In consequence, the steel tower becomes more sensitive to certain wind load conditions and experiences severe vibration frequently. To solve this problem, one type of steel–concrete hybrid wind turbine tower as shown in Figure 1, which is characterized by the lower part of the traditional steel tube tower replaced with the prestressed concrete segment, has been applied in bulk for Chinese wind farm construction from 2017. This kind of steel–concrete hybrid wind turbine tower combines the advantages of both steel tower and concrete tower. The upper steel part can take full advantage of easier prefabrication and assembling of steel structures. The bottom concrete part can lower the center of gravity of the tower and get a higher flexural stiffness than the full-height tubular steel tower. In addition, the concrete part can be prefabricated subassembly or casted in place according to the availability of concrete manufacturing and the project site. For most of steel–concrete hybrid wind turbine towers, their heights are usually higher than 100 m, and their costs have even amounted to around 30% of the total cost for wind turbine system. Therefore, construction cost is a key parameter that has to be considered in the design of steel–concrete hybrid wind turbine tower. Currently, the existing steel–concrete hybrid wind turbine tower has no significant advantage over the traditional steel tubular tower in construction cost. This is mainly because the design parameters including the geometrical shape, section dimensions, and heights of steel and concrete segments have not been optimized.

Optimization is the process of finding the optimal solution of a problem under one or several constraints. Some researches have been carried out on optimization methods and algorithms for solving optimization problems of structures, such as truss,^[16] steel frame,^[17] and steel lattice tower.^[18] The genetic algorithm (GA) was first developed by John Holland based on the Darwinian theory of evolution^[19] and then widely used by other researchers in complex problems. Its principles are encoding and improving complicated structures with simple representation and transformation. Galante^[20] applied GAs as a search technique to optimize trusses considering discrete design variables and market geometry values. Rajeev and Krishnamoorthy^[16] presented a GA-based methodology to solve size and configuration optimization problems design optimization of trusses. Erbatur et al.^[17] showed the efficiency of GAs as the optimizer for the discrete optimal design of steel frame and truss structures.



FIGURE 1 Steel–concrete hybrid wind turbine tower

Sivakumar et al.^[18] presented the optimization of steel lattice towers by combining GAs and an object-oriented approach. Kripakaran et al.^[21] presented a computational approach based on genetic algorithms to evaluate the cost of connections in the optimization of moment-resisting steel frames. Shen et al.^[22] used the GA to optimize the building frame column, in which a population initialization module is proposed to improve the optimization efficiency of the genetic algorithm. Baradaran and Madhkan^[23] presented an improved GA for the optimization of steel frame designs, in which the rate of convergence to the optimal solution was increased by splitting the population generation process to two stages. In this study, based on an improved genetic algorithm, a mathematical model is proposed for the structural optimization of steel-concrete hybrid wind turbine towers. In the optimization program, the design of the tower is optimized to reduce the construction cost and promote its application. Finally, the mathematical optimization model is applied to a 2-MW wind turbine to optimize the total cost of the support structure while satisfying multiple criteria imposed by design standards.

2 | MATHEMATICAL MODEL FOR OPTIMIZATION

2.1 | Design parameters

Generally, the typical structural optimization means to minimize the weight of a structure by adjusting design parameters such as lengths, diameters, and thicknesses of structural elements. However, the design optimization of wind turbine tower is not a simple task because of the wide variety of design parameters as well as the influence of external factors on the construction cost. The steel-concrete hybrid wind turbine tower was developed mainly to solve the excessive vibration and transportation difficulties of the pure steel tubular tower with the hub heights more than 100 m. Both the bottom concrete segment and the top steel segment are tubular and truncated cone shaped, in which the concrete segments are prefabricated modularly and assembled by means of posttensioning and the steel segments are connected by flanges in situ. Because of the cost difference of steel and concrete structures in material, fabrication, transportation, and erection, the dosage of both steel and concrete should have a proper proportion in order to obtain a minimum total construction cost. The design parameters including the outer diameter D_i , the thickness T_i , and the height H of the concrete part, the reinforcement area and prepressed force in the concrete part, are all considered as the optimization variables of the tower (shown in Figure 2), where D_1 and T_1 are the outer diameter and the thickness of the bottom end in the concrete part, respectively; D_2 and T_2 are the outer diameter and the thickness of the top end in the concrete part, respectively; $d_{s,b}(X)$ and T_3 are the outer diameter and the thickness of the bottom end in the steel part, respectively; D_4 and T_4 are the outer diameter and the thickness of the top end in the steel part, respectively. The concrete segment has a circular cross section whose diameter tapers linearly from the bottom end to the top end, as with the steel segment.

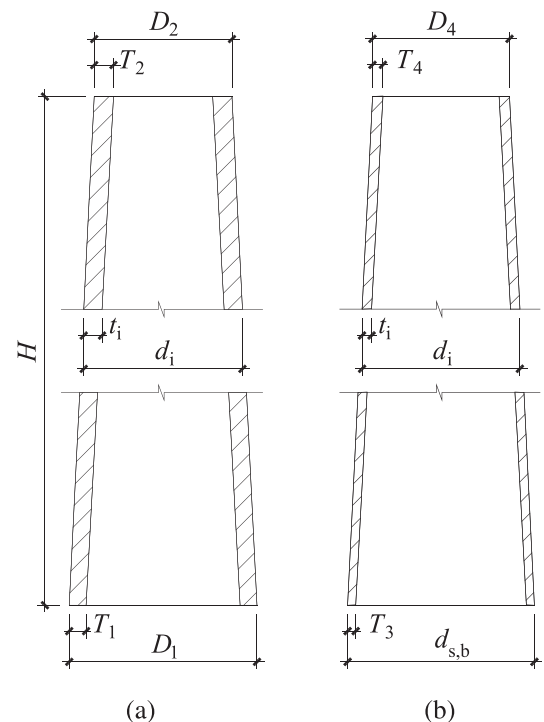


FIGURE 2 Design parameters of the tower: (a) concrete segment and (b) steel segment

2.2 | Basic assumptions and design conditions

The steel–concrete hybrid wind turbine tower is a typical cantilever beam built from segments with different cross-sectional properties. The structural model is developed using the linear elastic beam, in which its bottom end is assumed to be fixed to the ground and its top free end is attached a concentrated mass rigidly representing the inertia properties of the rotor and nacelle. All materials are supposed to be linearly elastic under the extreme load case and normal operational case. In the engineering practice, although the door opening weakens the concrete tube, some local strengthening measures are usually adopted around the door opening to reduce the stress concentration, such as setting steel door-frame and additional reinforcements. Therefore, the stress concentration around the door opening is not considered in the optimization model. The nonstructural mass including climbing aids, power transmission cables, safety ropes, vertical ladders with climbing protection (safety rope or safety rail), and platforms will not be optimized but included in the total mass of the tower during the optimization process. The nonstructural mass distribution along the height is taken equal to a fraction of the structural mass distribution (0.3). The pretension forces are applied on the concrete part to avoid the tensile stresses generated under the normal operational case and the cracks controlled to be the second level under the extreme load case (GB50010-2010).^[24] Considering that the prestressed force has almost no effect on the overall flexural stiffness of bonded or unbonded prestressed beams^[25] and the first vibration mode plays a dominant role for wind turbine system,^[26] the fundamental frequency and deflection of the tower are approximately calculated using the Rayleigh method by assuming as a generalized SDOF model.^[27] Considering the external forces acting in the fore-aft direction of the tower is much larger than that in the side-side direction, the optimization analysis is only carried out in the fore-aft direction.

2.3 | Optimization objectives

As the supporting structure of wind turbine system, an excellent design of the tower should have the optimal construction cost under the premise of ensuring the structural safety and functionality. Except for some factors such as fabrication, transportation, installation, and maintenance, the esthetic feature is also an important factor to determine the structural type of the tower for wind turbine owners. Generally, the objective function of this kind of optimization is set up by quantifying the degree by which the structure can succeed in achieving this objective by a single numerical value, such as the minimum weight, the maximum stiffness-to-mass ratio, or minimum vibration. In this study, the optimization work aims to minimize the construction cost of the steel–concrete hybrid tower by optimizing design variables. The mathematical description of the cost function can be expressed by Equation 1. It should be noted that a sufficient spacing between the natural frequency of the tower and the working frequency of the wind turbine should also be ensured to avoid resonance.

$$\begin{cases} \text{Minimize } f(X) \\ \text{Subjected to } X \in R, \\ R \subseteq U \end{cases} \quad (1)$$

where $X = [D_1, T_1, D_2, T_2, T_3, d_{s,b}, D_4, H]$ are the design variables, as shown in Figure 2; $f(X)$ is the objective function, which is the total cost of the tower considering all materials and the whole construction process, expressed as Equation (2); R is the set of optimization variables satisfying the design constraints; U is the fundamental space.

$$f(X) = V_{\text{con}}(X) \cdot p_{\text{con}}(X) + V_{\text{steelbar}}(X) \cdot p_{\text{steelbar}}(X) + V_{\text{pre}}(X) \cdot p_{\text{pre}}(X) + V_{\text{steel}}(X) \cdot p_{\text{steel}}(X), \quad (2)$$

where $V_{\text{con}}(X)$, $V_{\text{steelbar}}(X)$, $V_{\text{pre}}(X)$, and $V_{\text{steel}}(X)$ are the volume of concrete, steel bar, steel strand, and steel tower, respectively; $p_{\text{con}}(X)$, $p_{\text{steelbar}}(X)$, $p_{\text{pre}}(X)$, and $p_{\text{steel}}(X)$ are the comprehensive unit price of concrete, steel bar, steel strand, and steel tower, respectively.

2.4 | Design constraints

Currently, there are no special design specifications and standards for the steel–concrete hybrid wind turbine tower. The design constraints about the structural safety are established according to *Code for design of high-rising structures* GB 50135-2006,^[28] *Standard for design of steel structures* GB 50017-2017,^[29] and *Code for design of concrete structures* GB 50010-2010.^[24] The design constraints about the fundamental frequency are determined according to *Guideline for the certification of wind turbine* (GL 2010).^[30] Sixteen design constraints are shown in Equation (3), in which each inequality represents a specific constraint. The strength requirements under the extreme load case include the limit bearing capacity of the axial force and the bending moment in the concrete part and the limit strength, the equivalent stress considering the overall stability, and the equivalent stress considering the local buckling in the steel part. The strength requirements under the normal operational case include the vertical

tensile stress, the maximum principal stress, and the minimum principal stress in the concrete part. The horizontal displacement on the top of the tower under the normal operation case is calculated to ensure the clearance between the rotor rational plane and the tower. The fundamental frequency and the geometric shape of the tower, the minimum thickness of the concrete part, the dimension limits of the steel-to-concrete transition segment, the minimum inner diameter, and the maximum outer diameter are restricted as the additional constraints.

$$\left\{ \begin{array}{l} N_i(X) \leq R_{N,i}(X) \\ M_i(X) \leq R_{M,i}(X) \\ f_{s1,i}(X) \leq f_i(X) \\ f_{s2,i}(X) \leq f_i(X) \\ f_{s3,i}(X) \leq f_i(X) \\ \sigma_{ck,i}(X) - \sigma_{pc,i}(X) \leq 0 \\ \sigma_{tp,i}(X) \leq 0.85f_{tk} \\ \sigma_{cp,i}(X) \leq 0.60f_{ck} \\ u_{top}(X) \leq u_{allow} \\ f_{3P}/f_X \geq 1.05 \\ f_{1P}/f_X \leq 0.95 \\ d_{i+1}(X) \geq d_i(X) \\ t_{c,i}(X) \geq t_{c,allow}(X) \\ d_{s,b}(X) \geq D_2 - 2T_2 + 2\Delta + 2T_3 \\ d_i(X) - 2t_i(X) \geq D_{min,allow} \\ d_i(X) \leq D_{max,allow} \end{array} \right. , \quad (3)$$

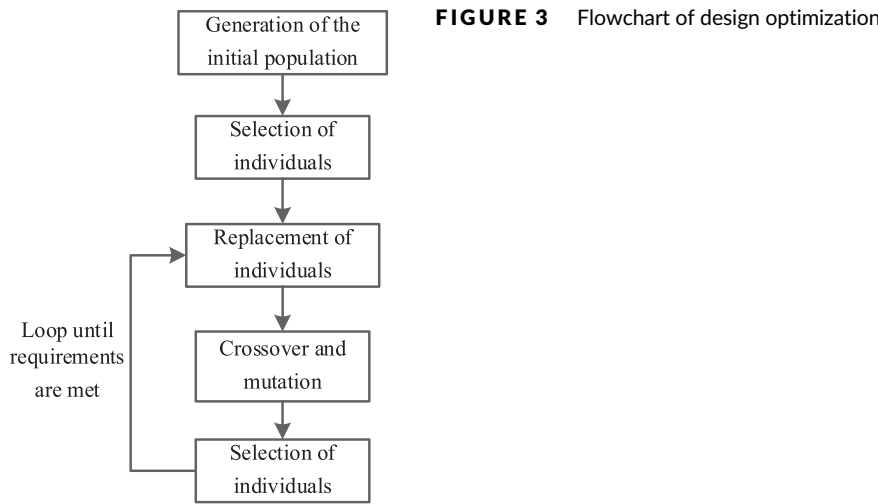
where $i = 1, 2, 3, \dots, n$, is the segment number from the bottom of the tower to its top; $N_i(X)$, $R_{N,i}(X)$, $M_i(X)$, and $R_{M,i}(X)$ are the axial force, the allowable axial force, the bending moment, and the allowable bending moment of the i th segment, respectively; $f_{s1,i}(X)$, $f_{s2,i}(X)$, $f_{s3,i}(X)$, and $f_i(X)$ are the design values of the maximum stress, the equivalent stress considering overall stability, the equivalent stress considering local buckling, and the design strength of steel in the i th segment, respectively; $\sigma_{ck,i}(X)$, $\sigma_{pc,i}(X)$, $\sigma_{tp,i}(X)$, $f_{tk}(X)$, $\sigma_{cp,i}(X)$, and $f_{ck,i}(X)$ are the design values of the stress, the prepressing stress, the maximum principal stress, the nominal value of tensile strength, the minimum principal stress, and the nominal value of compressive strength of concrete on the edge of i th segment, respectively; u_{top} and u_{allow} are the horizontal displacement and the allowable limit on the top of the tower, respectively; f_{1P} , f_{3P} , and f_X are the rotational frequency of the rotor, the frequency of passage of the rotor blades, and the fundamental frequency of the tower, respectively; $d_i(X)$ and $t_i(X)$ are the outer diameter and the thickness of the i th segment, respectively; $t_{c,i}(X)$ and $t_{c,allow}$ are the design thickness and the minimum allowable thickness of the concrete part, respectively; $d_{s,b}(X)$ and T_3 are the outer diameter and the thickness of the bottom end in the steel part, respectively; D_2 and T_2 are the outer diameter and the thickness of the top end in the concrete part, respectively; Δ is the minimum allowable distance between steel strands and the inner surface of the steel part.

3 | MATLAB PROGRAM FOR OPTIMIZATION DESIGN

The optimization goal is to find the near-optimal value of the objective function $f(X)$ in Equation 2 subjected to multiple constraints in Equation 3. The optimization process of the traditional genetic algorithm generally includes population generation, crossover, mutation, evaluation, and selection. In this study, the optimization algorithm is programmed based on the improved genetic algorithm, in which the replacement operator is introduced into the traditional genetic algorithm (shown in Figure 3). Individual genes are stored and operated using binary digits. The individual selections in subfunctions are operated using decimal digits such as the transformation of structural parameters, the calculation of structural capacities, and the checking of geometric parameters. The population size and the number of iterations are set to be 450 and 100, respectively.

3.1 | Optimization of initial population

The generation of the initial population needs three steps. First, in order to make the distribution of each initial individual in the solution space as wide as possible and then fully consider the possibility of various combinations of all design variables, the initial height of the concrete part (H) is determined according to the principle of the uniform distribution when the initial population is generated. Then, the initial values of other design variables are generated randomly. Finally, the generated individuals are checked to make sure whether they satisfy the constraints. If not, the



individuals will be regenerated until the number of individuals reaches the preset population size. The initial population will be improved until all design constraints of the individuals are satisfied. The flowchart of generating the initial population is shown in Figure 4.

3.2 | Replacement operator

To improve the premature phenomenon existing in the standard genetic algorithm, the replacement operator is introduced to solve the limitation induced by the early convergence and avoid getting a local optimization solution. In order to carry out the replacement operator, the genetic differences between adjacent individuals and between each individual and the optimal individual are calculated according to Equations 4 and 5, in which the genetic differences are represented by the vector differences between the binary genetic codes of the individuals. The individuals whose genetic differences are lower than the preset threshold will be replaced with the regenerated ones. And hence, the local optimal solution can be avoided by ensuring the genetic diversity of the population. The flowchart of the replacement operator is shown in Figure 5.

$$Difference_1(p) = \sqrt{\sum_{j=1}^m (Gene_p(j) - Gene_{best}(j))^2}, \quad (4)$$

$$Difference_2(p) = \sqrt{\sum_{j=1}^m (Gene_p(j) - Gene_{p+1}(j))^2}, \quad (5)$$

where $Difference_1(p)$ and $Difference_2(p)$ are the genetic differences between the p th individual and the optimal individual and between the p th individual and the next individual, respectively; $Gene_p(j)$, $Gene_{p+1}(j)$, and $Gene_{best}(j)$ are the j th genetic code of the p th individual, the $p+1$ th individual, and the optimal individual, respectively; m is the length of the genes.

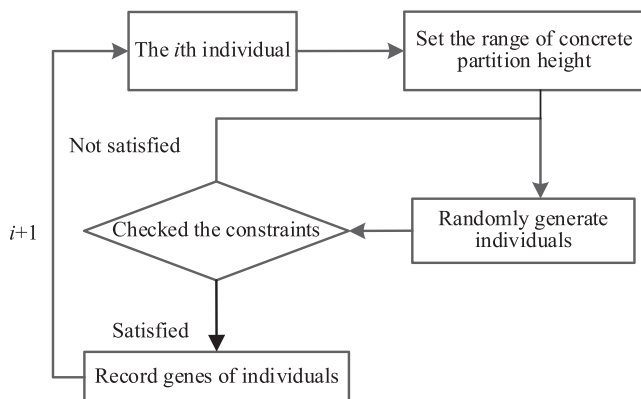
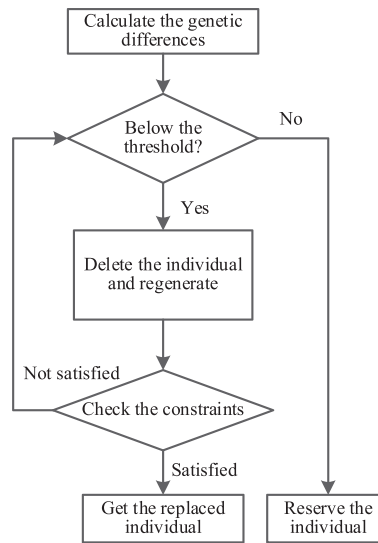


FIGURE 4 Flowchart of generating the initial population

FIGURE 5 Flowchart of replacement operator

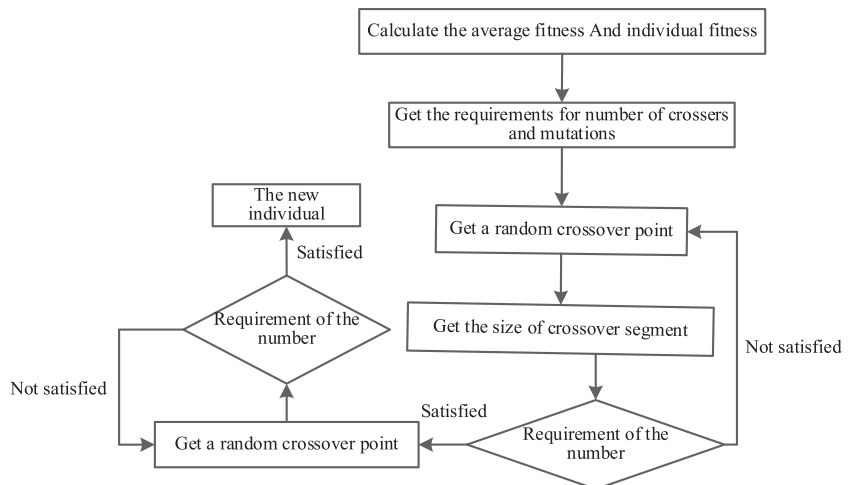
3.3 | Self-adaptive crossover and mutation operations

Crossover operator and mutation operator are two ways of producing new individuals with better fitness for genetic algorithm, as shown in Figure 6. The self-adaptive crossover and mutation operations with higher efficiencies are applied in this study.^[31] The number of crossover and mutation operations is automatically adjusted for each individual according to the relative relation between the individual fitness and the average fitness of the population. The individual with lower fitness value will experience a higher number of crossover and mutation operations. The number of self-adaptive crossover and mutation operations can be calculated by Equations (6) and (7).

$$n_c = \begin{cases} \frac{n_{cmax} - n_{cmin}}{1 + \exp\left(A \left(\frac{2(f - f_{avg})}{f_{max} - f_{avg}}\right)\right)} & f \geq f_{avg} \\ n_{cmax} & f < f_{avg} \end{cases}, \quad (6)$$

$$n_m = \begin{cases} \frac{n_{mmax} - n_{mmin}}{1 + \exp\left(A \left(\frac{2(f - f_{avg})}{f_{max} - f_{avg}}\right)\right)} & f \geq f_{avg} \\ n_{mmax} & f < f_{avg} \end{cases}, \quad (7)$$

where n_c is the number of crossover operations; n_{cmax} and n_{cmin} are the upper and lower limits of n_c , respectively; A is the adjustment coefficient,^[31] set to be 9.903; f , f_{avg} , and f_{max} are the individual fitness, the average fitness of the population, and the maximum individual fitness of the population; n_m is the number of mutation operations; n_{mmax} and n_{mmin} are the upper and lower limits of n_m , respectively.

**FIGURE 6** Flowchart of crossover and mutation operator

In this study, the upper limits of n_c and n_m are set to be 4 and 8, respectively; the lower limits of n_c and n_m are set to be 1 and 2, respectively. The relative relationship between the number of crossover and mutation operations and the individual fitness is shown in Figure 7, in which 0%, 50%, and 100% represent that the individual fitness is equal to the minimum, average, and maximum value of the population fitness, respectively.

4 | OPTIMIZATION OF A 120-M STEEL-CONCRETE HYBRID WIND TURBINE TOWER

In fact, the steel-concrete hybrid wind turbine tower is a competitive solution for the wind turbine tower taller than 100 m. Structural stability and durability due to cracking in concrete towers are a concern that further motivates the use of tall, hybrid towers.^[32] Hundreds of wind turbines with steel-concrete hybrid towers have been installed in China during the last 2 years (shown in Figure 1). The typical characteristics of this steel-concrete hybrid tower is composed of one prestressed concrete segment at its bottom and one traditional tubular steel segment on its top, which has better performance than concrete towers. In this paper, one market-leading steel-concrete hybrid tower with the height of 120 m^[33] for onshore wind turbine is optimized based on the proposed genetic algorithm. The optimization effect is evaluated by comparing with the original design.

4.1 | Design parameters

The rated power output of the wind turbine is 2.0 MW. The cut-in, rated, and cut-out wind speeds are 3, 8.6, and 19 m/s, respectively. The hub height of the wind turbine is 120 m. The concrete segment is 32 m, and the steel segment is 80.5 m in the original design. The diameter of the rotor is 115 m with a sweeping area of 10,381.6 m². The rotor and nacelle are simplified as a concentrated mass on the top of the tower. The total mass of the rotor and nacelle is 130×10^3 kg, and the center of the equivalent mass is located at the windward side deviating 3 m from the vertical axis of the tower. The size of the nacelle is $5.0 \times 4.5 \times 4.2$ m. The bottom segment of the tower is made of C60 concrete, and the top segment of the tower is made of Q345 steel. The steel bars in concrete are HRB400. The material properties of Q345 steel and C60 concrete are listed in Table 1. The diameter of single prestressing strand is 15.2 mm, and its reference ultimate strength is 1,860 MPa. The material properties are determined according to GB 50010-2010 and GB 50017-2017 and listed in Tables 1 and 2.

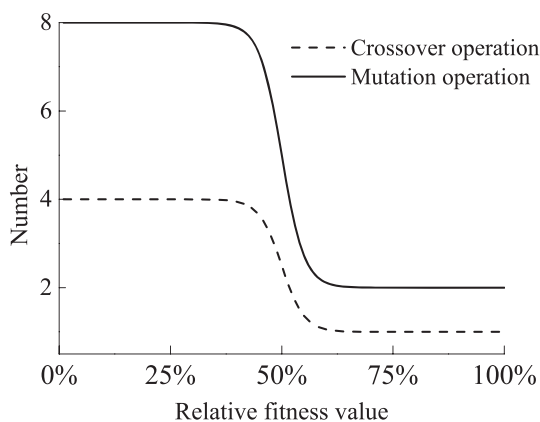


FIGURE 7 Number of self-adaptive crossover and mutation operations

TABLE 1 Material properties of structural steel, steel bar, and steel strand

Material property	Elastic modulus E_s (MPa)	Density ρ (kg·m ⁻³)	Poisson's ratio μ	Design strength f (MPa)
Q345	206,000	7,850	0.28	295
HRB400	200,000	7,850	0.30	360
Steel strand	195,000	7,850	0.30	1,320

TABLE 2 Material properties of concrete

Strength grade	Elastic modulus E_c (MPa)	Density ρ (kg·m ⁻³)	Poisson's ratio μ	Design value of tensile strength f_t (MPa)	Design value of compressive strength f_c (MPa)
C60	30,000	2,500	0.2	2.04	27.5

The wind turbine is built on open terrains, where the ground roughness should be classified as Class A according to *Chinese load code for the design of building structures* (GB50009-2012).^[34] The equivalent wind pressures corresponding the extreme wind speed and the wind speed under the normal operational case are 0.80 and 0.20 kN/m², respectively. The design service lives of the concrete segment are determined to be 50 years as a raised foundation according to FD 003-2007. The design service lives of the steel segment are determined to be 20 years according to *Guideline for the certification of wind turbines* (GL 2010).^[30] The reference wind pressure in the design service life is 0.4 kN/m². The wind loads acting on the tower are calculated according to GB50009-2012, and the parameters of wind velocity and wind profile are determined by GL 2010. The vertical force and the additional bending moment acting on the top of the tower induced by the weight of the nacelle, the blades, and the rotor are 1,300 kN and 3,900 kN-m, respectively.

The rotational speed of the rotor changes from 7 to 14.76 rpm, and then, the fundamental frequency of the tower should be greater than 0.259 Hz and less than 0.333 Hz, which is determined by upper limit of $f_{1p}/0.95$ and the lower limit of $f_{3p}/1.05$. The fatigue strength, deformation, and acceleration constraints are ignored during the optimization process and will be checked after an optimized solution is obtained. The self-weight of the tower is calculated automatically in MATLAB program by considering the dimensions of the tower and the material density.

4.2 | Cost estimation

The total construction cost of steel-concrete hybrid wind turbine tower is influenced by many complex factors such as the materials, fabrication, transportation, and installation method. For steel wind turbine tower, the evaluation method of its construction cost has been already very mature. But for concrete wind turbine tower, there is no accepted standard to estimate the formwork cost and installation cost because of the different fabrication mode and the installation conditions. The aim of this study is to give an optimal construction cost of steel-concrete hybrid wind turbine tower by the structural optimization. Therefore, a relative fair comparison is given to solve the lack of the evaluation criteria according to the civil construction and engineering experiences from the wind power industry providing the existing steel-concrete hybrid wind turbine towers.

At present, the total construction cost of one 120-m steel-concrete hybrid tower with 32-m concrete segment for 2.0-MW wind turbine is about 3.3 million RMB, in which the installation costs of the concrete segment and the steel segment are 190,000 and 200,000 RMB, respectively. The cost of one set of formwork for 32-m concrete segment is 2,600,000 RMB. But we should note, the formwork used for prefabricating the concrete segment can be recycled many times and then, one steel-concrete hybrid tower should share the formwork cost 52,000 RMB after 50 times recycling. The material unit price of Q345 steel is 10,000 RMB/ton, in which the fabrication and transportation costs have been included. Considering that the formwork and installation costs mainly depend on the height of concrete segment, the former is assumed to be proportional to the latter in this study. Accordingly, the comprehensive unit cost of the steel-concrete hybrid wind turbine tower is obtained and shown in Table 3.

4.3 | Optimization results

1 Cost optimization

The relationship between the total cost of the tower and the number of optimization iteration is shown in Figure 8. It can be seen that the total cost of the tower drops fast in the early stage of optimization, then slows down, and gradually converges in the later stage. Since the individuals are randomly generated in the initial population, the total cost in the first six optimization iterations is about 3.823 million RMB and higher than that of the original design. After the 13th iteration, the total cost is reduced to 3.3148 million RMB, which is appropriately equal to that of the original design. And then, the cost drops further and tends to be stable after the 350th iteration. The total cost is finally optimized to be 2.651 million RMB. That is to say, the total cost can be saved 0.6482 million RMB after optimization, which accounts for 19.64% of the total cost of the original design.

TABLE 3 Comprehensive unit cost of tower construction

Item	Unit price
C60 prefabricated concrete member	3,300 RMB/m ³
Steel strand	11,000 RMB/ton
Q345 steel	10,000 RMB/ton
Costs of formwork and installation for the concrete segment	7,563 RMB/m
Installation cost for the steel segment	2,273 RMB/m

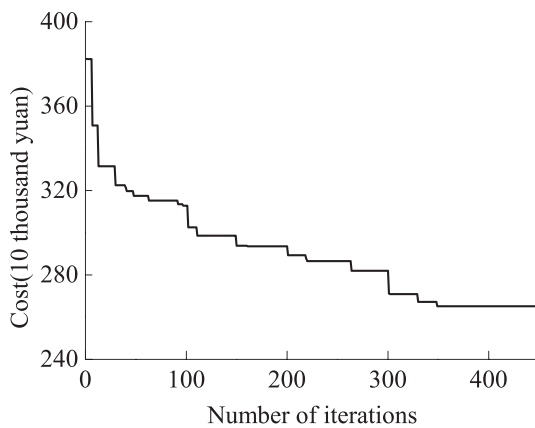


FIGURE 8 Cost convergence curve during optimization

2 Comparison of design parameters

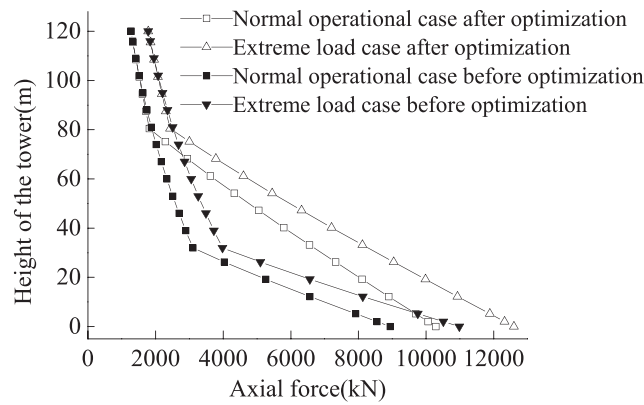
The comparison of the main design parameters for the steel–concrete hybrid wind turbine tower is shown in Table 4. It can be observed that the height of the concrete segment is relatively low before optimization. The construction costs of concrete and steel segments are about 1.2 million and 2.1 million RMB, respectively. After optimization, the height of the concrete segment is increased from 32 to 80.5 m. Accordingly, the flexural stiffness of the tower is also increased, which makes it easier to meet the requirement of the fundamental frequency and avoid the excessive vibration. In the meantime, the vertical compressive stress generated by gravity in the concrete segment is increased, which is helpful for steel strands to resist the vertical compressive stress induced by the bending moment. And then, the area of steel strands can be optimized, and the construction cost can be saved.

3 Comparison of the internal forces

The comparisons of axial forces in the tower before and after optimization under the normal operational case and the extreme load case are shown in Figure 9, from which some obvious changes can be observed. First, the height of the concrete segment in the tower after optimization is almost two and a half times before optimization. A significant turning point exists in the curves under these two load cases not only before optimization but also after optimization. After examining it carefully, the turning point is located at the joint between the steel segment and the concrete segment. This is mainly because the mutation of the axial stiffness occurs in this joint due to the changes of material density and section dimension. Second, the change range of axial forces along the height in the concrete segment is more obvious than those in the steel segment. The axial forces at the bottom end of the tower under the normal operational case and the extreme load case increase from 8,940.1 and

TABLE 4 Design parameters of steel–concrete hybrid wind turbine tower before and after optimization

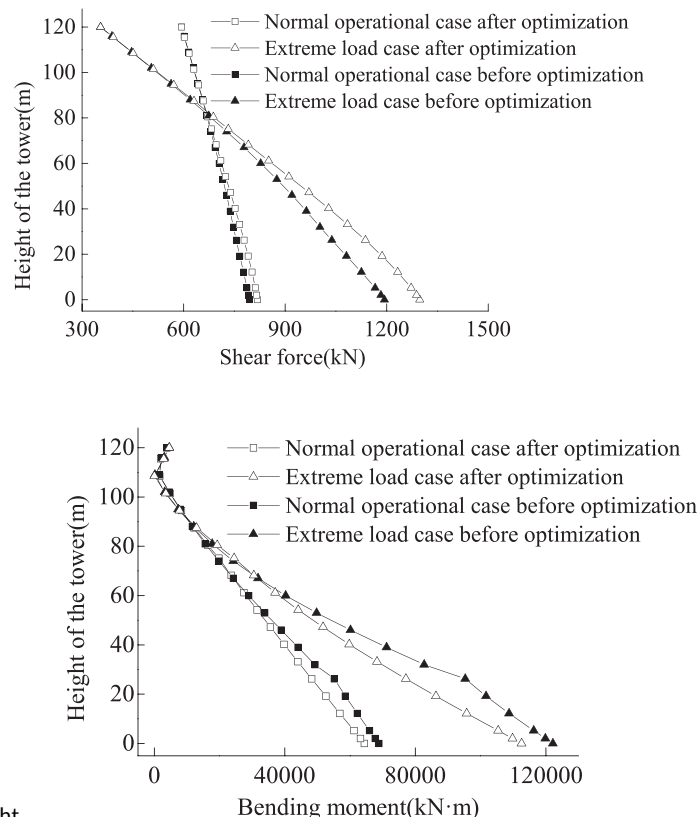
Tower	Parameters	Before optimization	After optimization
Concrete segment	H (m)	32.00	80.5
	D_2 (m)	4.86	3.90
	T_2 (m)	0.50	0.32
	D_1 (m)	7.76	6.96
	T_1 (m)	0.44	0.22
	V_{con} (m ³)	238.26	345.78
	V_{steelbar} (t)	10.62	15.41
	V_{pre} (t)	18.71	27.95
Steel segment	Height (m)	88.0	39.5
	D_4 (m)	3.28	3.22
	T_4 (m)	0.012	0.016
	$d_{\text{s,b}}$ (m)	4.30	3.87
	T_3 (m)	0.026	0.016
	V_{steel} (t)	186.61	55.05

FIGURE 9 Axial forces along the height

10,982.9 kN before optimization to 10,285.1 and 12,597.0 kN after optimization, respectively. This is mainly because the self-weight per meter along the height of the concrete segment is much larger than that of the steel segment.

Similarly, the comparison of shear forces in the tower before and after optimization under the normal operational case and the extreme load case is shown in Figure 10. Under the extreme load case, although the rotor is braked into a parked position, the horizontal concentrate forces acting at the top of the tower are lower than that under the normal operation case. However, the changes of shear forces along the height under the extreme load case are more obviously than that under the normal operational case because of the much larger wind speed. After optimization, the average diameter of the concrete segment is much larger than that of the steel tube at the same height before optimization. Thus, the windward area of the concrete segment increases after optimization and then leads to the increase of its shear forces. The maximum shear force at the bottom end of the tower increases by 8.73% from 1,193.6 to 1,297.8 kN after optimization.

The comparison of bending moments in the tower before and after optimization under the normal operational case and the extreme load case is shown in Figure 11. It should be noted that the bending moment caused by the eccentricity of the rotor and nacelle is opposite to that generated by wind loads. Therefore, an inflection point occurs at the height about 108 m of the tower. After optimization, the lateral deformation of the tower decreases as the increasing of its flexural stiffness, which leads to the decreasing of the additional bending moment caused by the concentrate mass on the top of the tower. Therefore, the bending moments along the height in the concrete segment decrease after optimization.

FIGURE 10 Shear force along the height**FIGURE 11** Bending moment along the height

Under the extreme load case, the bending moment at the bottom end of the optimized tower decreases by 7.85% from 122,136.6 to 112,543.3 kN-m.

4 Flexural bearing capacity

The flexural bearing capacities of the concrete segment along the height after optimization are calculated according to GB50135-2006^[28] and compared with that of the original design, as shown in Figure 12. Before optimization, the bending moments at the top and bottom ends in the concrete segment are 82,715.4 and 122,136.6 kN-m, respectively, while the flexural bearing capacities at the corresponding cross sections are 218,067.9 and 447,217.0 kN-m, respectively. The safety factors at the top and bottom ends of the concrete segment, which are defined as the ratios of the flexural bearing capacities to the bending moments, are 2.64 and 3.66, respectively. After optimization, the bending moments at the top and bottom ends of the concrete segment are 19,287.4 and 112,543.3 kN-m, respectively, while the flexural bearing capacities are 118,730.1 and 252,994.7 kN-m, respectively. The safety factors at the top and bottom ends of the concrete segment are 6.16 and 2.25, respectively. It should be noted that the safety factor at the top end of the concrete segment is larger due to the minimum workspace limitation of wind turbine system. However, it has little impact on the total construction cost since the diameter of this cross section is smaller. The safety factor of the flexural bearing capacity at the bottom end decreases after optimization, which is reasonable and could effectively reduce the total cost under the premise of ensuring the structural safety.

The checking results of the steel segment before and after optimization are shown in Figure 13. The strength, the equivalent stresses considering the overall stability, and the local buckling in the steel segment are checked according to the requirements of the large-diameter steel tube under the eccentric compression in GB 50017-2017^[29] and CECS 236-2008.^[35] Before optimization, the bottom end of the steel segment is lower, and its internal forces are larger. And then, the stresses in the wall of the steel tube are higher, and its safety factor is lower. After optimization, the height of the steel segment is greatly reduced, and the stresses in the wall of the steel tube are decreased. And then, the equivalent stresses considering both the overall stability and the local buckling are decreased, and the safety factors are improved accordingly.

5 Fundamental natural frequency of the tower

The Rayleigh method^[27] is used in this paper to calculate the fundamental natural frequency of the tower. The maximum rotating speed of the rotor is 14.76 rpm, and then, the corresponding rotational frequency is 0.246 Hz. Before optimization, the fundamental natural frequency of the tower is 0.286 Hz, which there is a safety margin of 0.042 away from the resonance frequency. After optimization, the fundamental natural

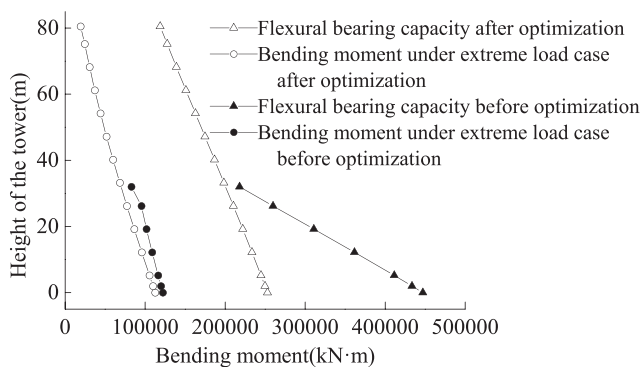


FIGURE 12 Flexural bearing capacity of the concrete segment

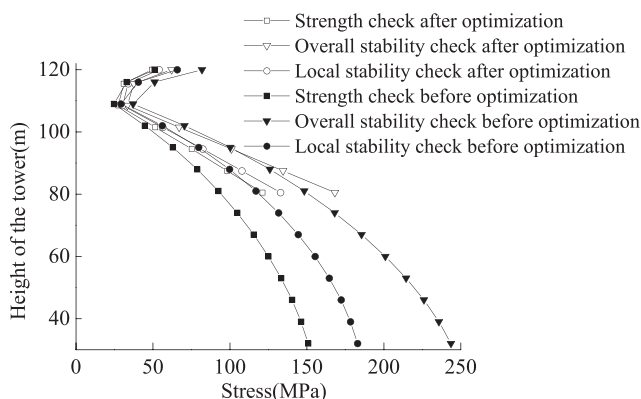


FIGURE 13 Stresses of the steel segment

frequency of the tower is 0.333 Hz, which there is a greater safety margin of 0.087 Hz away from the resonance frequency. This shows that the optimized tower has a better performance in resisting the excessive vibration because of its higher bending stiffness. The fundamental natural frequency both before and after optimization meets the relative requirements in GL 2010.

5 | CONCLUSIONS

The steel-concrete hybrid tower, which is characterized by a concrete tubular segment at the bottom and a steel tubular segment on the top, is a promising solution for ultrahigh wind turbine tower. In this study, a mathematical model is established to carry out the structural optimization of steel-concrete hybrid wind turbine tower. Influence factors of construction cost have been included in the proposed mathematical model, in which nonlinear multiple constraints are considered, such as constraint equations of stress limits, dimensional limits, and key operating parameters. The optimization results show that the optimal height of the concrete segment should be 80.5 m for one 120-m steel-concrete hybrid wind turbine tower. The total construction cost can be reduced about 20% after optimization. The optimization process shows that the genetic algorithm is robust and can meet the design requirements of preliminary design phase for steel-concrete hybrid wind turbine towers. After the height of the concrete segment is raised to 80.5 m, the fundamental frequency is increased effectively, and then, the optimized tower has a better performance in resisting the excessive vibration.

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